Reporting & Analysis Process (RAP)

Training for AQS Statisticians and AQS Statistical Programmers

Version 6
September 2025

NOVARTIS | Reimagining Medicine

Training Description

Purpose

This RAP WBT provides a detailed description of activities, roles, and responsibilities for conducting GxP Reporting Activities (RA) in a clinical trial, which are governed by these ESOPS:

- **SOP-7012383:** *Analyzing Clinical Data*
- **SOP-7017840:** *Programming Development and Validation in Statistical Computing Environments* and related documents (including templates)
    - **WP-7001131:** *Programming Development and Validation Instructions*
    - **GUID-8160112:** *Best Practices for Programming Development and Validation*
- **WP-8102363** *Establishing and Managing a Data Monitoring Committee (DMC)*

Scope

This WBT applies to all clinical trials in scope of the above-mentioned ESOPS.

Target audience

This WBT is relevant to AQS Statisticians and AQS Statistical Programmers supporting the in-scope RA.

Reminder – The content of this training does not replace the formal training on governing SOPs. Associates are expected to read and understand these SOPs and related documents.

Learning Objectives

Upon completing this training, you will understand the:

- **changes to the RAP** in this updated version
- **RAP documents/components** and their content **roles and responsibilities** of the RAP authors, reviewers, and contributors
- steps to **generate and maintain the RAP documents/components** and the end-to-end RAP process
- Interim Analysis (**IA)** process
- Data Monitoring Committee (**DMC)** process
- requirements for **other GxP RAs**

Key Updates in RAP WBT v6 (1/2)

- **General**
  - **Removed Medical Writer** role as in-scope for this WBT (i.e., MW will train on the 'RAP WBT for Non–AQS RAP Participants').
  - **Updated roles** to align with the current organization.
  - Lessons **content was re-grouped** to improve the flow of information and remove redundancy.
- **Lesson 1 CSR Activities**
  - **SAP template vs. WBT:** Overlap between the SAP template content and this WBT was addressed by retaining relevant information in the SAP template only and removing it from this WBT.
  - **TFL Shells:** Text was added that, as a best practice, TFL Shells development will begin after a content final SAP is available.
  - **SAP Addendum:** For post-DBL SAP changes which are not for primary or key secondary endpoints, the process was updated to include that the Trial Statistician (TS) / Independent Statistician (IS) ensures that SAP reviewers align on the changes added to CSR Appendix 16.1.9 and that this alignment is documented in the appropriate Meeting Minutes (GPT/GCT/CTT).

    Note: For post DBL changes to primary or key secondary endpoints, there is no change to the Addendum process (i.e., SAP Addendum is required.)

Key Updates in RAP WBT v6 (2/2)

- **Lesson 1 CSR Activities (cont.)**
  - **Medical Writer:** Removed as co-owner of CSR TFL Shells to align with SOP-7012383 *Analyzing Clinical Data.*
  - **Clinical Disclosure Office:** Specifications for disclosure reporting were removed and replaced with links to the relevant documents in go/CDO and go/CTSD. The CDO Registry Manager will review text in the SAP / TFL Shells related to the disclosure outputs.
  - **Companion Diagnostics (CDx) Bridging Studies:** Details on the roles required for the review and approval of the RAP documents/components were added.
- **Lesson 2 IA/DMC Activities**
  - **DMC:** Added information on the use of interactive visualization tools supporting DMC Closed Reports.
- **Lesson 3 Other GxP Reporting Activities**
  - Lesson renamed to emphasize **reporting activities are GxP.** Streamlined the content by removing what was already included in Lesson 1. Clarification added to the role names introduced and aligned responsibilities based on SOP-7017840 (and associated documents) and SOP-7012383.

Training Outline

Lesson 1: CSR Activities

RAP Documents / Components
RAP Roles & Responsibilities
RAP Document Management
Dry-Run Activities
Country Specific Analyses

RAP Documents / Components

Overview

Statistical Analysis Specifications

- **Statistical Analysis Plan (SAP)** contains a detailed description of the statistical aspects of a trial in alignment with the Protocol / Protocol Amendment(s). The Trial Statistician (TS) is author of the SAP and is responsible for the content and validity of the methodology to be applied.
- **Tables, Figures, Listings (TFL) Shells** document provides the mock-shells of the outputs supporting the planned analysis. The Trial Statistical Programmer (TsP) is the author of the TFL Shells document.

Technical Specifications and Tools

- **Programming Dataset Specifications (PDS)** contains all specifications (metadata and element derivations) to enable development /validation of derived-dataset programs, and review of metadata. The TsP is responsible for the content of the PDS.
- **Programming Deliverables Tracker (PDT)** lists all deliverables (outputs, datasets), agreed validation criteria, names of program developer / validator (as applicable), and other elements. The TsP is responsible for the content of the PDT, overseeing the programming activity.
- **Programming Summary Report (PSR)** details the RA validation strategy: it integrates the validation plan, and the PDT. It summarizes the programming verification and validation activities. The TsP is the author of the PSR.

Notes: 1) For the CSR, SOP-7017840 roles correspond to: Deliverable Analyst is the TS, Deliverable Lead is TsP, Deliverable Validator is a Programmer and/or a Statistician, and Deliverable Developer is a Programmer and/or a Statistician

2) All RAP documents are TMF-required but only the SAP is included in the Submission.

RAP Documents / Components

SAP

SAP – Purpose

- The SAP is a **scientific document,** and also a regulatory document. As part of CSR Appendix 16.1.9, the SAP is in-scope for public disclosure.
- The SAP provides **details of the statistical analyses** planned in the trial including 1) definitions of: hypothesis testing strategy, endpoints/estimands, analysis sets, derivations, analysis conventions; 2) handling of: intercurrent events, missing data, multiple records per visit, laboratory units. The SAP is aligned with the Protocol / Protocol Amendment(s) objectives, design, population, treatment, and assessments/procedures.
- The statistical analyses for all **primary and secondary objectives** are included in the SAP focusing on addressing Protocol questions.
- The SAP **does not include** any text which could lead to the revealing of commercially confidential or personally-identifiable information.
- The SAP template contains detailed information about the **expected content** of the SAP (e.g., analysis of endpoints and estimands, handling of intercurrent events, handling of missing data, analysis for pediatric studies, sensitivity analysis, and subgroup analysis).

SAP

RAP Documents / Components

SAP – Authoring Instructions (1/3)

- The **SAP content should be aligned** with the Protocol / Protocol Amendment(s) and data collection (i.e., CRF, Data Transfer Specification (DTS) from 3rd parties).
- **Model assumptions** should be written with sufficient details using plain language (descriptive text) to allow the correct programming of the model and the reproducibility of analyses.
  - If an installed software/package is to be used for analysis, to avoid unexpected results, ensure the intended method and model assumptions are correctly specified, using plain language and not as detailed lines of codes.
    - For example, in mixed models the Kenward-Roger (KR) method for approximating the denominator degrees of freedom is often recommended. However, in SAS PROC MIXED, it is not the default. The default is the “between-within” method.
- It is expected that Protocol requirements to **protect the blinding** are defined and implemented. If the Protocol does not include how the results will be disseminated, then this information must be included in the SAP. Refer to OneCTP template Section 6.3.2, with details provided in the Data Quality Plan (DQP).
  - There are no explicit external requirements/regulations in terms of how to disseminate study information during the conduct of the study, for example by treatment, in particular for blinded studies. It is expected that measures are implemented to protect the blinding.
- For **terminated studies**, if planned analyses cannot be performed due to data limitations, summary statistics at a minimum must be provided for primary and secondary efficacy endpoints, as appropriate, and rationale for not including the planned analyses should be included in the study report. Includes all definitions/criteria required to derive variables, e.g., primary / key endpoints. See go/CSR_Guidance.

RAP Documents / Components

SAP – Authoring Instructions (2/3)

SAP

- The SAP must be **finalized prior to DBL/unblinding** per SOP-7012383 *Analyzing Clinical Data.* However, it is **recommended** to **finalize the SAP prior to FPFV** (particularly for pivotal, open-label and single-arm trials) to provide evidence that analyses were planned before substantial data were available**.**
- Consider developing the **draft SAP** at a similar time as the statistical analysis sections of the Protocol.
- **Changes in primary / key secondary analyses / hypotheses** (including changes in testing hierarchy) are discouraged, unless there is a compelling reason (e.g., HA requests)
- In general, the statistical and clinical experts in the trial assess what **exploratory objectives** and/or exploratory analysis are intended to be included in the CSR. TS ensures the complete description is included in the SAP.
- Apply **SAP LEANing** principles and guidance on exploratory analyses (go/SMARThome).
- Use the **SAP template** and follow the template instructions as well as the OneCTP template instructions (e.g., blinding considerations, estimands (go/Estimands), etc.).

RAP Documents / Components

SAP

SAP – Authoring Instructions (3/3)

- Ensure that the SAP clearly **identifies the Protocol / Protocol Amendment version** that will be used (this was a HA Inspection finding) and documents any changes and rationale for the updates.
- Ensure that the SAP clearly defines the RA that will be using the **SAP version.** Decide whether for DMC / IA, separate analysis plans will be created.
- The SAP should be written in a way that allows **efficient translation** into the TFL Shells and PDS. Methodology/definitions must be described completely in the SAP, not in the TFL Shells or PDS.
- The SAP should use the **standard language** in the SAP template, guidance documents and disease/therapeutic Standards in go/RAP, when available.
  - These standards should be **copied into the SAP** rather than being referenced, so that a stand-alone document can be sent to HAs, if needed.
- **Outputs required for disclosure** should be specified in the SAP as defined in: "SAP & TFL Considerations for Data Disclosure of AE & All-cause mortality data" which is available in go/CTSD under "Planification". Additionally, a guidance document entitled "CDO Guide to the Reporting & Analysis Process (RAP)" provides disclosure requirements relevant to the TS. This document is owned by CDO and is located in go/CDO. For study related questions, please contact "Registries, TrialAndResults (Gen)" <trialandresults.registries@novartis.com>"
  - The **CDO manager** will review the SAP text / TFL Shells related to the disclosure outputs (safety XMLs files).

RAP Documents / Components

TFL Shells – Content (1/3)

- TFL Shells document is to be approved by 6 months after FPFV (best practice, see slide 33). As a best practice, TFL Shell development will begin after a content final SAP is available. This will help to ensure that late changes to the TFL Shells are minimized.
- TFL Shells contain details of the layout of all tables/figures/ listings, including titles, headers, programming specifications and footnotes.
- TFL Shells are developed based on the following documents: Protocol, SAP, Case Report Forms (eCRF), Data Transfer Specifications (DTS) for 3rd party transfers. Each of these documents provides key information for the correct display of the TFL Shells for the intended analyses:
  - The Protocol delivers key elements related to the design/treatment such as treatment arms / cohorts, schedule of assessments/visits,
  - The SAP provides detailed description of the endpoints (e.g., efficacy, safety, PK), estimands, analysis, analysis sets, definition of baseline, treatment periods, derivations (if needed), timepoints and corresponding windows to be applied.
  - eCRF provides the data types, code-lists, etc.
  - The DTS provides details of what will be captured from 3rd party data, e.g., domains, parameters, blinding, masking.
- TFL Shells should be consistent with the SAP version used for the analysis. The TFL Shell document should include the SAP version, as indicated in the template.

RAP Documents / Components

TFL Shells – Content (2/3)

- The design of the outputs (i.e., TFL Shells) should **enable effective interpretation** of the data in line with the objectives.
- TFL Shells should be based on available **standards** (e.g., global standard TFL library, indication standards). Shells that fit the intended analysis should be selected from the current library. If there is no available TFL standard or no shell with the same type of analysis, a request to the Clinical Data Standards Governance Board should be made.
    - A cross-functional agreement for the new requirements is needed before submitting a request.
    - It is recommended to add any new shells to the standard library for use on future studies.
- The **eRAP** tool (or equivalent tool approved for usage) should be used when preparing TFL Shells.
- TsP to ensure TFL Shells **suitable for use** within the CSR body are included**.**
- If results from an **exploratory objective** are agreed to be included in the CSR, the TFL Shells document needs to be updated accordingly.
- TFL Shells should not include anything which could lead to **disclosure** of commercially confidential or personally-identifiable information.
- TFL Shells should **include all unique shells**, including standard shells, to facilitate internal and external (e.g., Contract Research Organization (CRO) / HAs) review of planned outputs.
- If the same TFL Shells document is used for **multiple analysis time points,** then it should include which outputs are generated for each analysis time point (e.g., DMC, interim, primary, etc.).

RAP Documents / Components

TFL Shells – Content (3/3)

TFL Shells

- There are different **types of analyses and considerations** including (but not limited to) the following:
  - **Frequency** tables – it should be clear which categories are to be summarized and how missing values are displayed (denominator = patients with non-missing data or all patients in the analysis set?)
  - **Summary** tables of continuous data – are percentiles needed or only min-max, mean (std) and median? How to display (e.g., number of decimals)?
  - **Inferential** tables, e.g., analyses of continuous data – model should be clearly described, or time-to-event data – standard Kaplan-Meier shells used
  - Mandatory outputs related to clinical **disclosure** (EudraCT, CT.gov, EU CTIS).
- **Footnotes** may be included to aid understanding of the TFL Shells, especially if the same output is to be used for different Reporting Activities; however, the footnotes **should not include detailed statistical methodology** or programming specifications, rather use keywords/few lines to add key information about the methodology.
- For **complex statistical models** (e.g., Bayesian models), the TS provides the appropriate TFL Shells /layout and the TsP provides the TFL Shells numbering.

RAP Documents / Components

TFL Shells – Authoring Instructions (1/2)

- Ensure **titles** are rendered correctly when no population is selected for in-text outputs.
- Tables should not display **blanks**.
- Agree on the method for calculation of **percentiles**.
- The ICH E3 output **numbering convention** for TFL Shells should be followed.
- Refer only to **variables/parameters/fields** in TFL Shells that are captured in the database (unless for derived parameters).
- **Standard shells** that display subgroups side by side are available, consider using these for some domains, e.g., Adverse Events (AE).
- In the global library, most of the table shells include a "**total" column**. Assess whether a 'total' should be displayed or not in a particular table.
- For **IA or long-term trials**, it is recommended that TFL Shells be final by 6 months prior to IA date/cut-off date.

RAP Documents / Components

TFL Shells – Authoring Instructions (2/2)

- For **terminated studies**, if planned analyses cannot be performed due to data limitations, all primary and secondary (both efficacy and safety) endpoints should be summarized in a CSR table. If there is >1 patient in the trial, the number of TFLs can be reduced, a summary table must be created for disposition, demographics, and each primary and secondary outcome measure. Exploratory endpoints can be omitted altogether.
- **Color graphs** can be used as long as the graphics are still distinguishable in shades of black and white. Refer to go/IRDDS link Section 2.13.1, for detailed requirements.

RAP Documents / Components

PDS

PDS - Content

- In the **current GPS environment**, the PDS is an excel file containing metadata along with derivations in accordance with the SAP / TFL Shells. PDS documents are utilized to manage dataset, programming development and validation activities.
- **PDS includes**
  - All variables required for the reporting needs as defined in the SAP / TFL Shells.
  - The derivation rules for each variable and the corresponding metadata (i.e., type, format, length and label, etc.)
  - Imputation rules

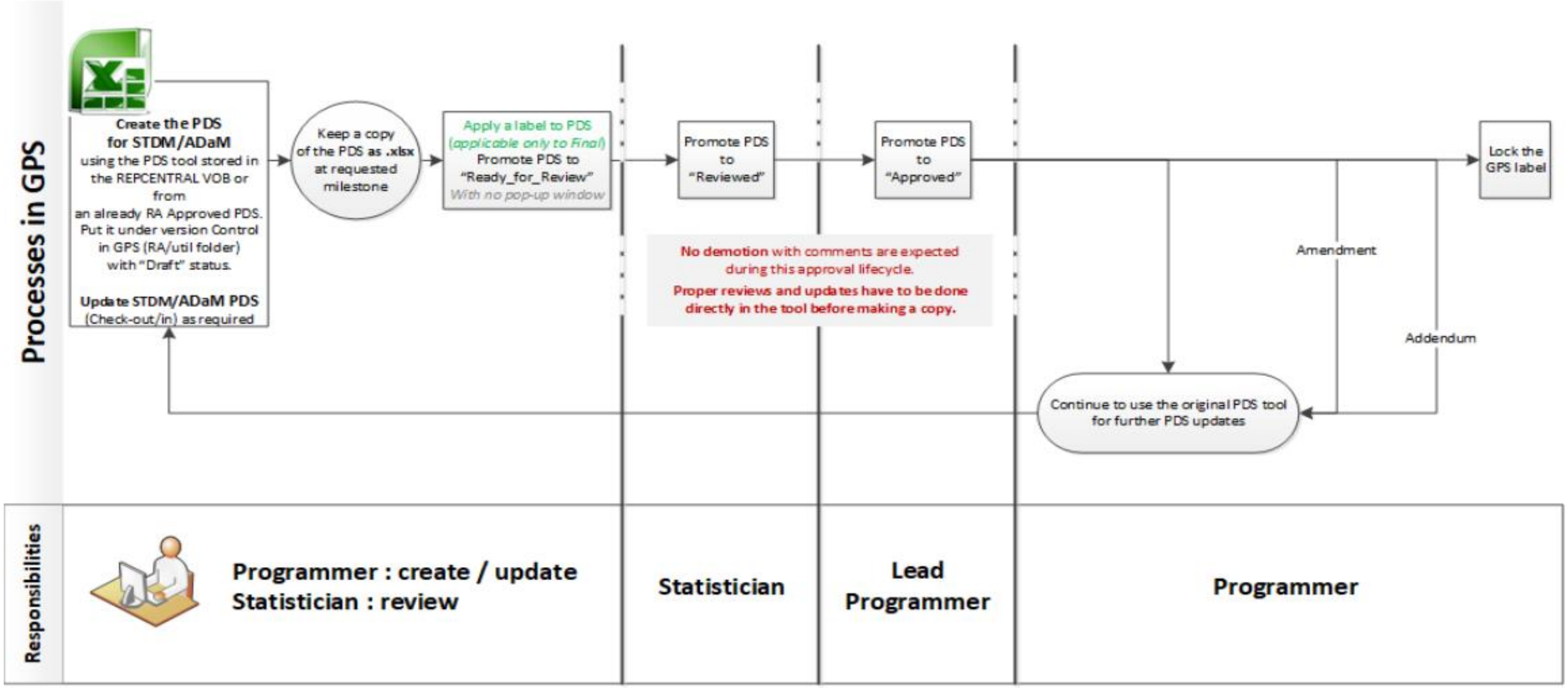
  Note: Refer to the Roadmap (slide 34) for the best practice timelines for finalizing the PDS.

- TsP is responsible for copying the template ***PDS.xlsm*** from the central location to the correct util folder, which may differ depending on the GPS release.
- TsP ensures that all variables to be reviewed by the TS are flagged in the column "Require Statistician Review". TsP/TS to agree on what variables are relevant .
- TsP ensures all standard derivations are clearly defined, where required.
- TsP ensures that the PDS must be aligned with the SAP & TFL Shells.
- TS reviews the PDS to ensure the methodology and definitions are correctly implemented.
- TsP promotes (in the Lifecycle management) PDS.**xlsx** and not PDS.xlsm.

PDS

RAP Documents / Components

PDS – Content Workflow and Responsibilities



Source: More details are provided in (PRADA) Programming Datasets Specifications (PDS) Workflow

RAP Documents / Components

PDT - Content

PDT

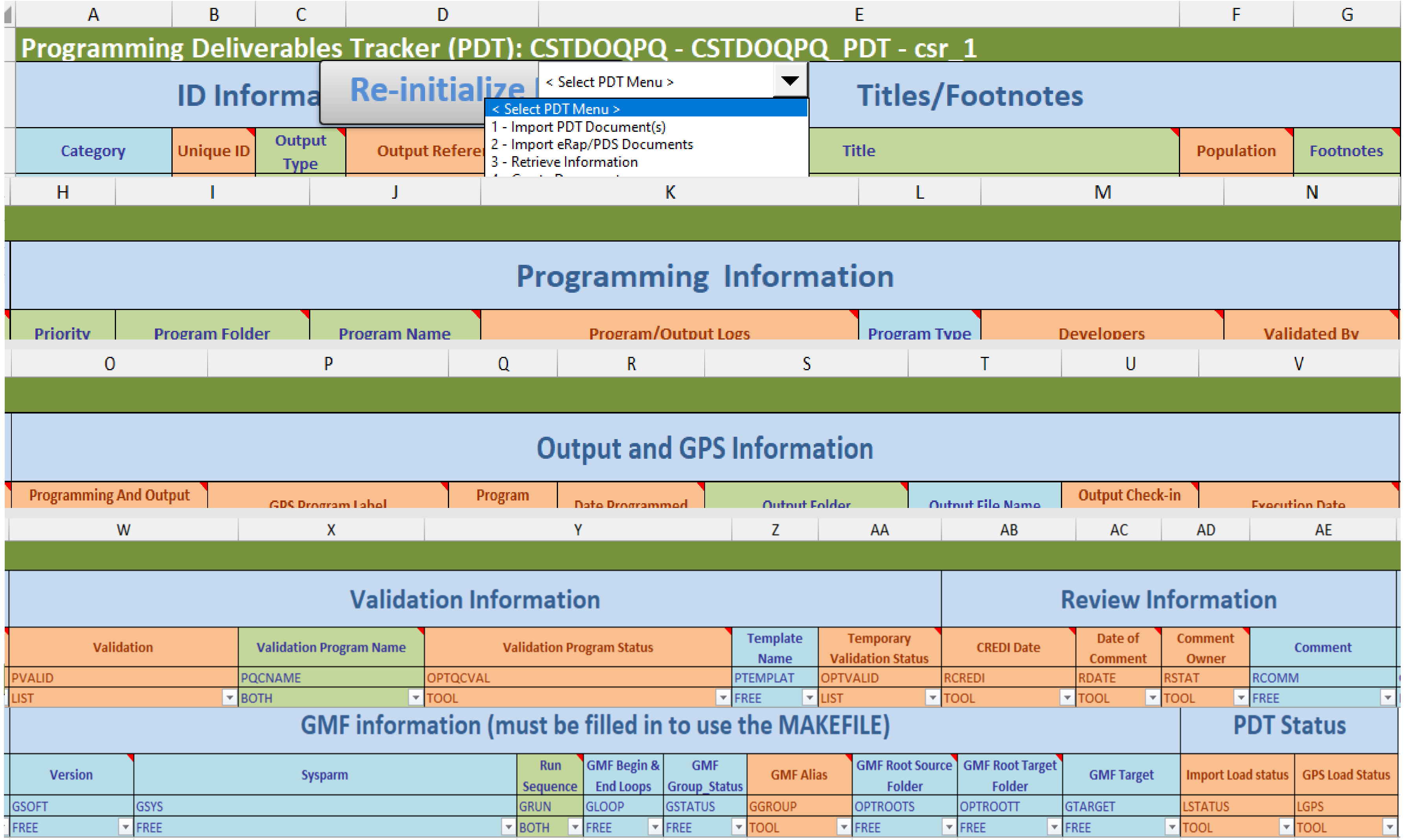
In the **current GPS environment**, PDT is a tool used to track programming activities reflecting the validation plan for the RA.

- PDT has the **functionality to read the TFL Shells** elements obtained from the eRAP.
- All information on datasets, tables, listings, figures and their attributes are **captured in the PDT.**
- The file **extension ".xlsx"** is used for PDT lifecycle promotions while referring to it in the PSR.
- The PDT must be **aligned with the validation plan**. The document is promoted to be «reviewed» by the TS in the Statistical Computing Environment (SCE) before starting the programming activity.
- The PDT file is locked following the assignment of the "Reviewed" status (for identification in any version tree). To update a "Reviewed" PDT, the TsP **needs to demote the file.**
- **Tracking of deliverables** using the PDT is initiated with the validation plan, per process. TsP/TS review and documented the tracker in GPS.
- **TsP/TS review the PDT** before any deliverable is released, to ensure it is aligned with the validation plan for any changes in the TFL Shells.
- The PDT should be **finalized** before CSR publication in DMS. If finalized thereafter, it is considered as an Addendum and the filename should reflect this.

RAP Documents / Components

PDT - Components

PDT



RAP Documents / Components

PSR

PSR - Content

- The PSR is a document that **integrates the validation plan and the PDT**, and summarizes the programming verification and validation activities performed for a RA. In addition, the PSR
  - details information relative to the SCE used for the RA
  - includes a section to detail any findings in production logs, including corrections and the risk assessment (of the finding).
- The PSR should be **initiated before the start of programming activities** to ensure it captures the methods/techniques to be used as validation strategy, (e.g., code review/ programming, code testing). If a technique other than code review/programming is used for validation, the PSR details the expected outcome to consider the validation completed successfully.
  - The TsP, in collaboration with the TS, establishes/updates the validation strategy, following the process in SOP-7017840. This first version of the PSR is documented in the SCE before start of programming activities.
- The PSR details **inconsistencies in production outputs,** for which an update/re-run of programs is needed. In addition, it includes the impact assessment (of the finding), and details of corrections made.
- The PSR should be finalized before CSR publication in DMS. If finalized thereafter, it is considered as an Addendum and the filename should reflect this.

Back to TOC

Lesson 1: CSR Activities

RAP Documents / Components
RAP Roles & Responsibilities
RAP Document Management
Dry-Run Activities
Country Specific Analyses

RAP Roles & Responsibilities

Role Name Conventions by Organization[1]

Role Name	Role in BR: TCO / TM / CP	Role in Development	Role in GMA
Lead Statistician (LS)	Project Statistician[2]	Lead Statistician / Global Program Biostatistics Head	Lead Statistician
Lead Programmer (LP)	Lead Programmer / Project Programmer[2]	Program Programmer / Project Programmer	Lead Programmer
Medical Lead (ML)	**TCO**: Clinical Program Lead **TM**: Translational Medicine Expert **CP studies**: • CP Medical Lead (fully outsourced) • CD Medical Director or TCO/TM roles above (in-house)	NA (This naming convention is not applicable to Development since there is a Medical Lead but the RAP support is primary from the CD Lead.)	Medical Lead
Study Lead	**TCO**: Clinical Science Trial Leader **TM**: Clinical Scientist **CP studies:** • CP Study Lead (fully outsourced) • Global Trial Director or TCO/TM roles above (in-house)	NA (There are no tasks for the Development Study Lead in this WBT.)	NA (There are no tasks for the GMA Study Lead in this WBT.)

Notes:
[1] Only roles which are named differently in BR, Development and GMA are listed here. (TS, TsP, and MW are not listed here since they are named the same in all organizations.)
[2] In BR, Project Statistician (PS) roles can be taken by TS or PS; Lead Programmer / Project Programmer roles can be taken by the next higher level Line Function level.
Acronyms are listed/decoded in the Acronym list at the end of this document.

RAP Roles & Responsibilities

Role Definitions

- **Author** is the content developer. There is a single primary Author for each RAP document.
- **Contributor** provides information and may draft certain sections (e.g., Safety, Pharmacokinetics and Pharmacodynamics (PKPD)).
- **Reviewer** reviews the content for correctness based on expertise.
  - Reviewers do not necessarily approve the document in the system.
- **RAP participants (Contributors/Reviewers)** includes Statistician (Trial/Lead), Statistical Programmer (Trial/Lead), Medical Writer, CD Lead (Development), and Medical Lead (BR/GMA).
  - Other experts including relevant external partners may participate as Contributor, Reviewer (e.g., Safety Lead, Study Lead (BR), Pharmacometrics Modeler, Pharmacokinetic Sciences (PKS) Lead, Biomarker Expert, Patient-Reported Outcomes (PRO) Expert, Imaging Expert, etc.). For CDO reviewer info, see slide 12. For CDx Bridging Study participants, refer to slide 27.
  - Depending on the study, additional internal Contributor/Reviewer may approve the SAP and/or TFL Shells.
- **Approver** approves the content in the appropriate tool/system. For documents governed by SOP-7012383 and SOP-7017840, the document author must ensure that:
  - The approver(s) is an independent person(s) from the Author (SOP-7018147 *GxP document Lifecycle Management)*.
  - Approval can be delegated appropriately **within the same LF** to the next higher function level, where applicable.
  - Depending on the study design, other contributors approve the SAP / TFL Shells documents, e.g., PK specialist.

RAP Roles & Responsibilities

Authors, Participants and Approvers for the Original Document*

BR / Development / GMA Trials

RAP Document / Component	Author	Participants (Contributors / Reviewers)	Approvers
SAP	TS	All RAP participants**	LS, TsP, MW, ML(BR/GMA), CD lead (Dev), PKS Lead (if PK is primary endpoint) **Note:** Other approvers may be added based on the study specifics.
TFL Shells	TsP	**Contributor:** TS (other Contributors based on study specifics) **Reviewers:** All RAP participants*	TS, LP, MW, ML(BR/GMA), CD lead (Dev), PKS Lead (if PK is primary endpoint) **Note:** Other approvers may be added based on the study specifics.
PDS (SDTM & ADaM)	TsP	**Contributors:** Program Developers, Program Reviewers **Reviewer:** TS	LP
PDT	TsP	**Contributors:** Program Developers, Program Reviewers **Reviewer:** TS	*No Approval*
PSR	TsP	**Reviewer:** TS	LP

Notes: *For Amendments and Addendums, refer to slide 38.
**Definition of RAP Participants is provided on the previous slide.

RAP Roles & Responsibilities

Authors, Reviewers and Approvers

CDx Bridging Studies

RAP Document / Component	Author	Reviewers	Approvers for Original Versions, Amendments and Addendums
SAP	TS	**Reviewers*:** Precision Medicine (PM) team, Biomarker Trial Manager, TsP, LS	LS, PM Lead and/or Diagnostic Sciences Lead
Sample selection memo (if applicable)	TS	**Reviewers*:** PM team, Biomarker Trial Manager, TsP, LS	LS, PM Lead and/or Diagnostic Sciences Lead
TFL Shells	TsP	**Reviewers*:** PM team, Biomarker Trial Manager, TS	TS, PM Lead and/or Diagnostic Sciences Lead.
PDS (SDTM & ADaM)	TsP	**Reviewer:** TS	LP
PDT	TsP	**Reviewer:** TS	*No Approval*
PSR	TsP	**Reviewer:** TS	LP

Notes:
CDx Bridging Studies are described in go/CDx.
SAP and TFL Shells are separate documents.
* Representatives from the External Diagnostic device company may also be Reviewers, if applicable

RAP Roles & Responsibilities

Outsourced studies using CPP* Model

- External partners (e.g., CROs) must use **Novartis templates** (e.g., SAP, TFL Shells, CSR) and follow applicable guidance documents, unless otherwise specified in the contract agreement.
- The **same timelines** for SAP and TFL Shells finalization apply as in-house studies, unless otherwise specified in the contract agreement.
- **Primary Author** of RAP document(s) are external partners, unless otherwise specified in the contract agreement.
- **Novartis TS will load the SAP** into the DMS and promote for approval after team review and CRO addresses comments; Clinical Pharmacology (CP) Study Lead will load the TFL Shells into DMS and promote for approval after team review and CRO addresses comments. The **approvers** are the same as for in-house trials except TP and MW are not needed (MW is not assigned to these trials). The final approved documents in DMS should be shared with the external partner.
- **Novartis TsP sets up study folders** in the SCE and secure platform to support data exchange between Novartis and external partners (e.g., Secure Vault) for the trial. Review access on the exchange is provided to CRO programmers.
- **Novartis TsP** ensures that all **programming deliverables** (SDTM and ADaM datasets, TFL Shells, PDS, PDT, programs, Pinnacle21 reports, etc.) are correctly finalized in DMS or SCE, per process.
- For **additional information** regarding CPP model studies, see go/CP or contact the CP Subject Matter Experts (SME) listed in Acknowledgments slide.

Note: *Clinical Pharmacology Partnership where CROs provide services from Protocol writing to CSR finalization

RAP Roles & Responsibilities
Functional Outsourced Studies

- External partners (e.g., CROs) should use **Novartis templates** (e.g., SAP, TFL Shells) and follow guidance documents (CSR listings, etc.) as agreed in the Study Specific Worksheet (SSW). Note, for example, Research Collaborations may use their templates.
- The **same timelines** apply as in-house studies, unless otherwise agreed in the contract agreement.
- If the SAP is prepared in-house and the programming is outsourced, the TS should ensure that SAP **comments from the external partners** are received/reviewed via the agreed-to process.
- If the TFL Shells, PDS or PDT are prepared in-house, the TsP should ensure that **comments from the external partners to the programming RAP documents** (TFL Shells, PDS, PDT, as appropriate) are received/reviewed via the agreed-to process.
- If the SAP is outsourced, **the Primary Author** of RAP document(s) should be the external partner, unless otherwise specified in the contract agreement.
- If external partners do not have access to Novartis DMS, Novartis TS and TsP are **responsible for uploading outsourced documents**, ensuring all documentation is complete (e.g., TS loads the SAP; TsP loads the TFL Shells).
  - Novartis TsP sets up study folders in SCE and the secure platform to support data exchange between Novartis and external partners (e.g., Secure Vault) for the trial. Review access on the exchange is provided to CRO programmers.
  - Novartis TsP ensures that all programming deliverables (SDTM and ADaM datasets, TFLs, PDS, PDT, programs, Pinnacle21 reports, etc.) are correctly finalized in DMS or SCE, per process
  - The SAP review/approval/finalization process is aligned with the in-house trial. (Note: If the vendor follows their own SAP review/approval process, the Novartis TS loads the SAP into DMS and the approvers includes at a minimum the GPBH/LS.)
- The final documents in DMS and the SCE should be shared with the external partner. List of DMS Approvers and date of approval may be obtained from DMS and shared with external partner, if needed.

Note: For **additional information** regarding Outsourced studies see go/Analytics_Outsourcing

Back to TOC

Lesson 1: CSR Activities

RAP Documents/Components
RAP Roles & Responsibilities
RAP Document Management
Dry-Run Activities
Country Specific Analyses

RAP Document Management

Repositories for TMF-required Documents

The **SAP, TFL Shells and PSR** documents should be created using the most recent templates available in the Novartis Clinical Vault (NCV).

- **SAP:** Statistics › Statistics Oversight › Statistical Analysis Plan; then Choose template: SAP (Statistical Analysis Plan)
- **TFL Shells:** Statistics › Statistics Oversight › Statistical Analysis Plan; then Choose template: TFL Shells
- **PSR**: "Statistics > Analysis > Analysis QC Documentation", then Choose template: PSR (Programming Summary Report)

The **PDS and PDT** should be created using the most recent templates in the SCE. For example, in GPS:

- **PDS:** REPCENTRAL\RAP_Templates\PDS\PDS.xlsm
- **PDT:** REPCENTRAL\RAP_Templates\PDT\PDT.xlsm

RAP Document Management

Naming Conventions

Naming convention for the RAP documents

<*Trial number*>_<Document acronym>_<*Reporting Activity name*>_<#>_<*optional additional RAP identifier*>:

- ‘Trial number’ must include the prefix “C”
- ‘Document acronym’ must be included after trial number. ‘Document acronym’ could be SAP, TFL, PDS, PDT or PSR
- ‘*Reporting Activity name*’ and _<#>’ where <#> represents a consecutive number which ideally follows GPS / Computing Environment folder structure, see generic examples and QVM GPS Structure Example below
- ‘*optional additional RAP identifier*’ used to add further clarity

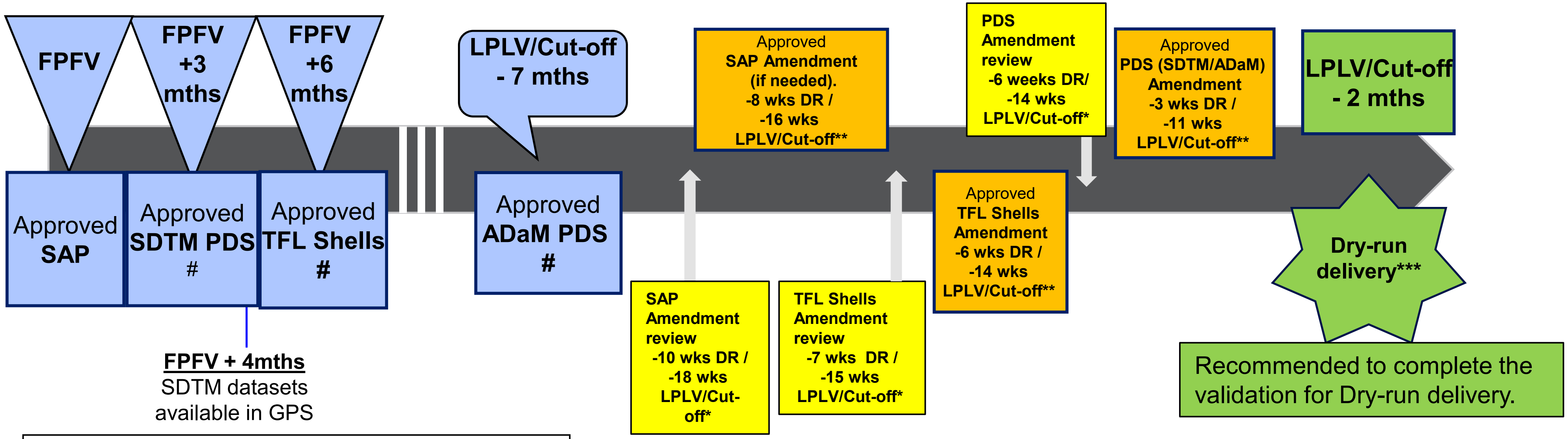
- For Amendment: _amendment<n>, where n=1,2...
- For Addendum: _addendum<n>, where n=1,2...

Examples:

- SAP,
  - CABC123D4567_SAP_CSR_1
  - CABC123D4567_SAP_CSR_1_primary
  - CABC123D4567_SAP_CSR_2_final
  - CABC123D4567_SAP_CSR_1_amendment1
- TFL Shells,
  - CABC123D4567_TFLS_CSR_1
  - CABC123D4567_TFLS_CSR_1_primary
  - CABC123D4567_TFLS_CSR_2_final
  - CABC123D4567_TFLS_CSR_1_amendment1
- SDTM-PDS,
  - CABC123D4567_PDSSDTM_CSR_1_final
  - CABC123D4567_PDSSDTM_CSR_1_amendment1
- ADaM-PDS,
  - CABC123D4567_PDSADAM_CSR_1_final
  - CABC123D4567_PDSADAM_CSR_1_amendment1
- PDT,
  - CABC123D4567_PDT_CSR_1_final
- PSR,
  - CABC123D4567_PSR_CSR_1

RAP Documents Management

Finalization Timelines Roadmap, Best Practices up to Dry–run



Note:
For trials

- with short duration < 1 year or
- with IA shortly after FPFV or
- in Early Development AQS (ED-AQS) or
- with new or complex indications or
- where HA interactions may lead to additional SAP changes,

the timelines may be adjusted accordingly.
These timelines are not applicable for Clinical Data Complete (CDC) Dry-run.

***New review timelines (applicable only if amendment is needed)**

****New approval timelines (applicable only if amendment is needed)**

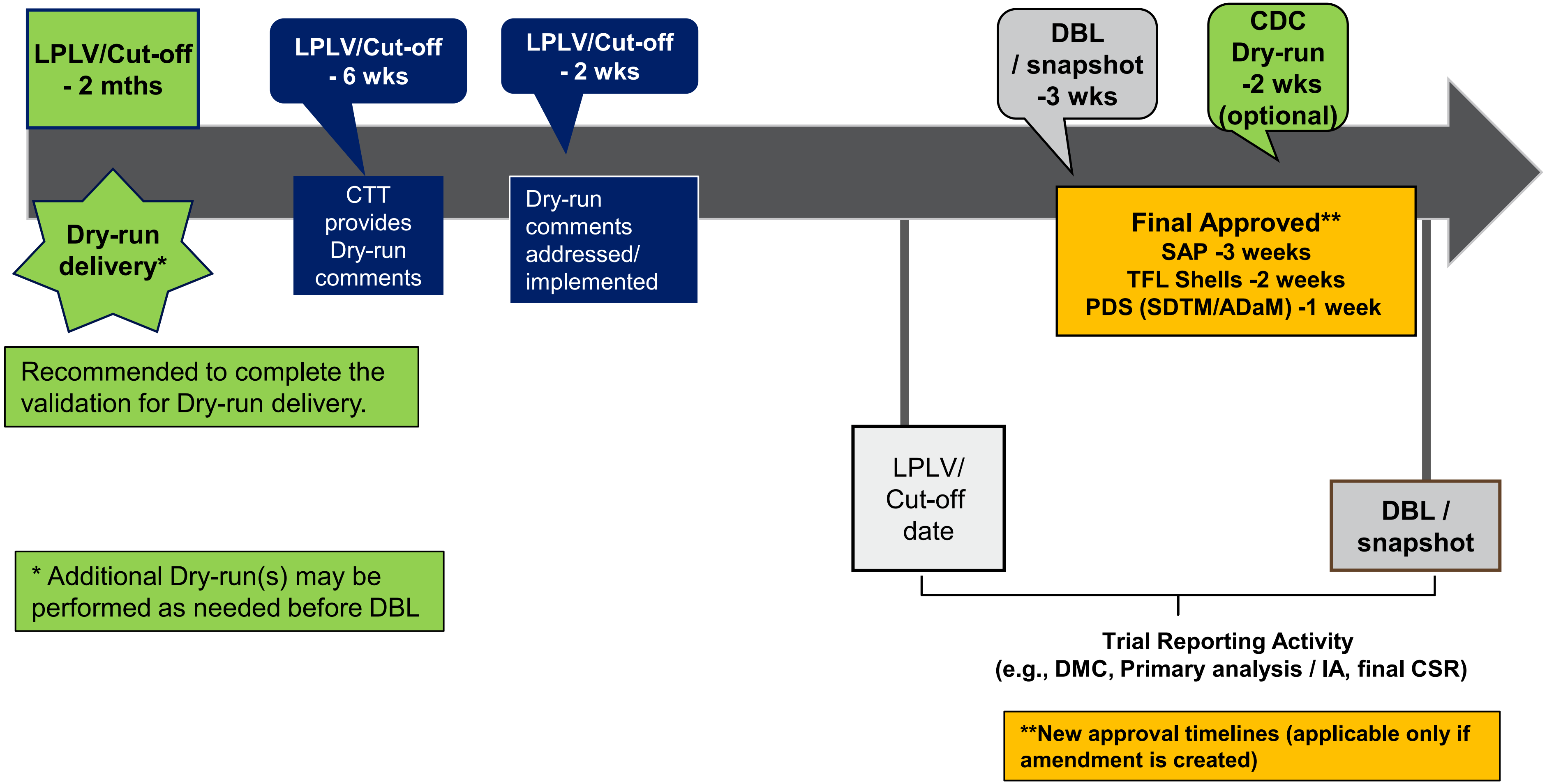
*** Additional Dry-run(s) may be performed as needed before DBL

 # Before starting the programming activities, align the validation strategy with TS in the PSR. And the PDT must be promoted to 'reviewed' by the TS for the criticality levels alignment as per PSR.

Notes: GPS = Global Programming and Statistical Environment
DR = Dry-run

RAP Documents Management

Finalization Timelines Roadmap, Best Practices for DBL



RAP Document Management

Amendments / Addendums (1/3)

- **Changes in planned analyses** are any changes to the analyses specified in the Protocol (or amended Protocol). Describe these changes in **CSR Section 9.8.3.** and indicate whether they occurred before or after DBL/unblinding.
- **Changes** to the specifications (SAP, TFL Shells, and PDS) **after DBL should be avoided.** Changes to the SAP / TFL Shells may trigger updates to the PDS. The TS and TsP should ensure these 3 documents are aligned.
- Ideally there should be **no SAP Amendments or Addendums** but in the case of **compelling scientific justification** (e.g., new external evidence), an Amendment or Addendum may be necessary.
- If required, the SAP Amendment should be **finalized as soon as possible.**
  - Carefully consider the content of the SAP Amendment. Depending on the **importance of a particular change** (e.g., to change the approach to control the Type I error), a Protocol Amendment may be needed, and the SAP will then need to be updated accordingly. In some cases (e.g., minor updates to the analyses set or a change to an exploratory analyses) amending the SAP only is appropriate.
- **Amendments: Changes before DBL:** Updates to statistical methods or planned analyses prior to DBL/unblinding must be made in the SAP and reflected in the TFL Shells and PDS, and the CSR / outputs.
- **Addendums*:* Changes after DBL:**
  - **SAP:** For any changes to the **primary or key secondary endpoint analyses** (e.g., HA requests), **documentation must be included in a SAP Addendum**, review must be performed by all SAP reviewers (slide 25) and approval by roles listed on slide 38. (Also, see Appendix example.)
    - TS (or IS) to ensure that the document history of the SAP Addendum includes a high-level description of changes and the justification for these changes. In addition, TS (or IS) to ensure alignment with Lead Statistician.
  - **SAP:** For changes to any **other endpoints or additional analyses,** the TS (or IS) should:
    - **Update CSR Appendix 16.1.9** with the changes. [Note: TsP will create TFL Shells Addendum.]
    - **Ensure all SAP appropriate reviewers** (slide 25) review and agree to the changes. (Note: The reviewers may be CTT equivalent roles at the GPT/GCT level.)
    - **Finalize** the Appendix **in DMS** prior to the final run of the additional outputs.
    - At the next scheduled **CTT Meeting or GPT/GCT Meeting (as appropriate),** document in the Minutes that the changes were reviewed by all relevant SAP reviewers. Note: The Minutes should capture the roles performing the review and agreeing to the changes but **not details of the analyses.** This will protect the blinding, when relevant.
  - Documentation **of other RAP document** changes: Update as an Addendum using the Addendum process including approvals (slide 38).

RAP Document Management

Amendments / Addendums (2/3)

- **Maintenance of the SAP and TFL Shells** throughout the trial is a critical task. It ensures that any changes to analyses are transparently-documented and that the impact of any such decisions on the quality, the integrity of the overall trial analysis, and resources, is assessed and agreed upon before changes are implemented. All changes should be captured by updating the SAP, TFL Shells, and/or CSR, as appropriate.
- **Changes** to the SAP and TFL Shells **may be driven by several factors,** including:
    - the availability of **new information** regarding the drug (which may be influenced by results from other studies), or drug class (perhaps a new safety concern)
    - in response to specific **HA feedback**
    - a **Protocol Amendment** (e.g., with changes relevant to trial design or data collection)
    - the need to **clarify any technical specifications** once programming on live data begins (e.g., to adapt for any unexpected data scenarios)
    - changes in **disease-area standards**
    - **compound/trial terminated** for futility/safety or other reasons (see guidance on needs for abbreviated/close-out CSR, go/SMARThome in the Process Recalibration site)

RAP Document Management

Amendments / Addendums (3/3)

- The **Data Exploration Strategy (DES)** document (go/SMARThome) specifies the exploratory analyses that are planned outside of the SAP. If the CTT intends to report any results from the DES in the CSR, then the SAP and/or TFL Shells must be updated accordingly.
- **Rationale for changes** should be summarized in the RAP documents Change History Table.
- It is recommended to **collate multiple changes** to the SAP / TFL Shells before finalization of SAP / TFL Shells amendment prior to Dry-run / DBL / milestone to minimize the number of amendments.
- As a minimum, RAP approvers should be **informed of any intended changes** in the analysis plan and reach an agreement.
- **Late changes may delay deliverables**! Timelines should be reviewed and if necessary, updated.
- **Early engagement** of all parties in the SAP and TFL Shells development process should reduce the need for changes.
- The **impact of inconsistencies** between Protocol, SAP, TFL Shells and outputs identified after the published CSR should be assessed by the CTT and Line Function (LF) Managers and, documented in the CTT meeting minutes.
- Note: **A tracked-change version of the SAP is not required. However, some HA may request this version.**
    - If created, it is a best practice to store it in the DMS without initiating the workflow. See go/AQS_NCV_FAQs for details on creating/archiving the tracked-change version.

RAP Document Management

Amendments / Addendums Approval

Development / GMA Trials

RAP Document/ Component	Amendment Approvers	Addendum Approvers
SAP*	LS, TsP, MW, ML(GMA), CD Lead (Dev), PKS Lead (if PK is primary endpoint)	AQS DU Head, DU CD Head (Dev); GMA Global Head DU (GMA)
TFL Shells	TS, LP, MW, ML(GMA), CD Lead (Dev), PKS Lead (if PK is primary endpoint)	LS, LP, MW, ML(GMA), CD Lead (Dev), PKS Lead (if PK is primary endpoint)
PDS (STDM & ADaM)	LP	LP
PDT	*No Approval*	*No Approval*
PSR	If change occurs after PSR is final and before CSR is published, then Approver is LP (same as for original PSR)	If change occurs after PSR is final and after CSR is published, then Approver is LP (same as for original PSR)

BR Trials

RAP Document/ Component	Amendment Approvers	Addendum Approvers
SAP*	LS, TsP, MW, ML, PKS Lead (if PK is primary endpoint)	CP studies: ED-AQS GGH, PKS TA/DU Head TCO: ED-AQS GGH, TCO Global Head TM: ED-AQS GGH, TM DU Head
TFL Shells	TS, LP, MW, ML, PKS Lead (if PK is primary endpoint)	LS, LP, MW, ML, PKS Lead (if PK is primary endpoint)
PDS (SDTM & ADaM)	LP	LP
PDT	*No Approval*	*No Approval*
PSR	If change occurs after final PSR and before CSR is published, then Approver is LP (same as for original PSR)	If change occurs after final PSR and after CSR is published, then Approver is LP (same as for original PSR)

Notes: DU = Development Unit; TA = Therapeutic Area

* For changes to the primary or key secondary endpoints, the SAP Addendum is to be created and approved as indicated. For changes to other endpoints, the CSR 16.1.9 is to be used and the TS is the approver. See slide 35.

For CDx Bridging Studies' RAP document Amendments and Addendums Approvers, see slide 27.

Back to TOC

Lesson 1: CSR Activities

RAP Documents/Components
RAP Roles & Responsibilities
RAP Documents Management
Dry-Run Activities
Country Specific Analyses

Dry-run Activities

Process

Per SOP-7012383 *Analyzing Clinical Data*, the CTT should **assess the need** for a Dry-run. This section provides guidance for studies which include a Dry-run.

- The **purpose** of the generation and review of the Dry-run is to:
  - **Ensure that outputs are consistent** with the specification's documents and full coverage of the SAP at predefined milestones before a production run.
  - **Build-in and affirm quality early** in the analysis process to help ensure programming and validation steps are on schedule.
- **The Dry-run deliverables are not designed, and should not be used, as prospective data cleaning** tools or to track the progress of the external delivery (i.e., outsourcing).
- The Dry-run process is **led by the TsP.**
- At **least one Dry-run** is recommended before DBL, additional optional Dry-runs may be performed.
- The **data cutoff**, percentage of clean patients (if applicable) and display of real treatments, etc. should be discussed and aligned within the CTT.
  - A Dry-run is optional for subsequent data cut-off(s).
- For blinded trials, **dummy treatment codes** must be used.

Dry-run Activities
Dry-run Review

- Dry-run TFLs should be **reviewed in the appropriate system** (e.g., DMS).
- **All RAP participants** perform detailed review of Dry-run outputs to ensure these match specifications.
- The recommended **Dry-run reviewers** are TS, MW, CD Lead (Development), Medical Lead (BR/GMA), Study Lead (BR), and Safety Lead.
- Other experts and relevant external partners may **contribute to the review** (e.g., Biomarker Expert, Patient-Reported Outcomes (PRO) Expert, Imaging Expert, Modeler, etc.).
- Reviewers enter comments, in the **Dry-run Comment Sheet** (or directly into the computing environment, if available).
- If needed, a **Dry-run review meeting** can be conducted by TsP to resolve comments and to agree on the responses.
- Based on the **review of comments**,
  - TsP to resolve programming issues
  - TS and TsP to update appropriate RAP documents, if required.
- If **systematic issues** are identified through the programming process, these should be communicated and addressed in the CTT with Data Management (DM) and Clinical, and their detection should be incorporated in the appropriate data cleaning process, independently of the Dry-run.

Back to TOC

Lesson 1: CSR Activities

RAP Documents/Components
RAP Roles & Responsibilities
RAP Document Management
Dry-Run Activities
Country-Specific Analyses

Country-Specific Analyses

- **Country representatives** (e.g., Japan statistician) should be involved in developing the RAP of global studies so any country needs can be considered in the trial SAP (e.g., definitions/rules, additional Adverse Event (AE) tables by time period, etc.).
- For submissions in some countries/regions, not only **separate Clinical Pharmacology (CP) studies** might be required, but also **subgroup analyses** of the respective country/region (e.g., Japan or East Asia) patients included in the, usually pivotal trial.
- The selection of **TFL Shells for the respective subgroup** (country/region) should be indicated in the SAP only if Regulatory Affairs (RA) confirms it needs to be pre-specified in the SAP and/or included in the main CSR (or in a separate country-specific SAP, if deemed necessary).
    - The selection of TFL Shells and further specifications (e.g., not to display p-values, different titles/footnotes, etc.) should be **provided by the country representative(s)**
- It should also be clearly stated if the **country-specific TFL Shells** are included in the main CSR or described separately (e.g., First Interpretable Results (FIR) for decision making and/or 'abbreviated CSR' used for New Drug Application (NDA) (to be discussed as per country needs)).
- **For Japan HA**, the CSR-TFL Shells do not satisfy the needed requirements by the Japan HA (PMDA) for submission activities. Please refer to WP-7007641: JPTO_*JOM-00105_Guidance for how to handle Japan specific outputs for Trial/Project Level. Also, s*ee guidance in go/SMARThome.

Back to TOC

Lesson 2: Interim Analyses / DMC Process

NOVARTIS | Reimagining Medicine

Interim Analyses (1/2)

- IAs are any analyses that take place **while a trial is ongoing** and prior to final database cleaning and locking. This broad definition is included in the FDA guidance "Adaptive Designs for Clinical Trials of Drug and Biologics" (FDA, 2019):
  - *"An interim analysis is any examination of data obtained from subjects in a trial while that trial is ongoing and is not restricted to cases in which there are formal between-group comparisons."*
- IAs may be based on:
  - Unblinded data (presented by treatment whether coded or not) as per the ICH E9 definition of an IA: "Any analysis intended to compare treatment arms with respect to efficacy or safety at any time prior to the formal completion of a trial."
    - These IAs may **be carried out for safety or efficacy** assessments (e.g., monitoring of safety data or futility analysis for early stopping or formal primary analysis for HA submission). Often unblinded IAs are reviewed by a DMC to preserve trial integrity or blinded data investigating nuisance parameters (e.g., variability of an important endpoint).

Interim Analyses (2/2)

- There are **two common types of IAs** at Novartis:
  - IAs for DMCs - details are provided in this Lesson
  - IAs for CSRs - the CSR process described in Lesson 1 applies
- An Independent Statistician (**IS**) and/or an Independent Programmer **(IP**) is involved for blinded trials, as applicable. It is also possible that a blinded and an unblinded team is necessary. The CTT will perform a risk assessment and decide what is more appropriate to control the blinding.
- If an IA **was not planned** in the Protocol (including Amendments), then it would need approval following the process described in SOP-7012383 *Analyzing Clinical Data.* For the approval requirements, refer to ‘Approval Form for Unplanned Interim Analysis’ in DMS.
- The following IAs **are not in scope** for this Lesson:
  - “Preliminary” analyses (e.g., SOP-7017621: *Handling of Bioanalytical Data for Clinical Trials*).
  - ED-AQS Protocol-defined exploratory IA performed to generate data insights.
  - Non-randomized Safety only or dose escalation IAs.

Back to TOC

DMC Process

Introduction

- A Data Monitoring Committee (DMC) is a group of **independent experts** that reviews the ongoing conduct of a clinical trial and the emerging data to ensure **continuing patient safety**, as well as the **validity and scientific merit** of the trial.
- A DMC may cover a **single trial or several trials** within a program.
- A DMC can also be referred to as an Independent DMC (**IDMC**) or a Data Safety Monitoring Board (**DSMB**).
- DMCs make recommendations based on **reviews of unblinded Interim Analysis (IA) data**.
- ***References and Resources***:
    - SOP-7012382: *Establishment and Management of Data Monitoring Committees (DMCs), Adjudication Committees (ACs) and Steering committees (SCs)*
    - WP-8102362: Establishing and managing a Data Monitoring Committee (DMC)
    - DMC Charter template and DMC Charter Signature template in DMS
    - GUID-8170268: *DMC Closed Reports*
    - GUID-8170974: *Internal Guidance on Data Monitoring Committees*
    - go/DMC
    - go/dmc/report

DMC Process
Overview

- The use of a DMC should be **pre-planned** and described in the Protocol.
- The **DMC Charter** defines the scope of data review and issues to be addressed by the DMC. The Charter must be finalized before FPFV in the study part covered by the DMC (e.g., core). The DMC Charter template (in DMS) lists the approvers.
- A **DMC Closed Report** is a well-structured, easily-navigable, stand-alone document describing the unblinded IA data. It is designed to support decision-making during the course of the trial in the context of incomplete and inconsistent data.
  - Teams are encouraged to create a DMC Closed Report which is not just a collection of outputs. This Report is only accessible to the DMC during the conduct of the trial.
  - Details on how to structure a DMC Closed Report are available in GUID-8170268: *DMC Closed Reports* .
  - An example of a DMC Closed Report is available at go/DMC.
- DMCs should have **access to all data domains** (safety, labs, efficacy data, demographics, etc)
  - The relevance of particular data domains should be discussed and agreed with the DMC.
  - Access to efficacy data should be provided even when there is no preplanned IA for efficacy.
- Data may **not need to be fully cleaned** at the time of the DMC meeting. Limitations in data quality should be communicated to all reviewers of the data (e.g., DMC members). The Data Quality Team (DQT) should define the cleaning strategy and document it in the Data Quality Plan (DQP).

DMC Process
Trial Integrity

- DMCs review IAs based on unblinded data. As such **protecting trial integrity** is paramount and analyses for DMCs are usually created by persons outside of the clinical team.
- For **confirmatory studies** or studies that might be supportive during a submission**,** analyses for DMC review should be created by an external analysis center, e.g., CRO.
- If the **IS and IP are from Novartis:**
  - Outputs for DMC Reports should be created in a **controlled area** within SCE (e.g., a restricted folder in GPS) and the TsP should ensure that the IS/IP have appropriate access to the controlled area.
  - Note: "For some BR trials where the TS/TsP are unblinded to treatment information per Protocol, they may perform the IS/IP activities.
  - The IS and IP are responsible for completing the training in Up4Growth.
- If the DMC Closed Report **masks the treatment names** (e.g., uses labels such as ‘A’, ‘B’), the DMC Chair should receive the decoding prior to the first DMC meeting.
- The TS will request the **release of the randomization data** from the Randomization Office (RO) (go/Randomization_Office) to authorized recipients. The TS will not have access to any unblinding information, this will only be available to the IS and the IP.

DMC Process

SAP, TFL Shells, PDS, PDT and PSR

- **Author** of SAP and TFL Shells:
    - Trial Statistician or Lead Statistician
- **Approvers** of SAP and TFL Shells (including Amendments):
    - Lead Statistician
    - Trial Statistical Programmer
    - CD Lead (Development)
    - ML (BR/GMA)
- For each DMC review based on an IA, there should be a **separate PDS** (only when there are changes from the previous analysis), **PDT and PSR**.
    - For DMCs, the PSR is a live document in SCE under study UTIL, and for each DMC, a new version is created. After the last DMC or at DBL, the **final PSR version** should be approved and archived into DMS, per process. For more information refer the PSR template.
- **Following the first DMC review**, if subsequent review requirements are fully covered by the previous documentation, then the previous SAP, TFL Shells and PDS can be referred to and no further SAP, TFL Shells or PDS are needed.

DMC Process
Documentation

- Usually specific DMC SAP, TFL Shells and PDS are created. However, the **main RAP documents** can be referred to or used for this purpose, if analyses for the DMC are clearly specified and the data displayed appropriately.
- In addition to the “standard” review by the key RAP participants and the IS and IP, the DMC SAP and TFL Shells should be **discussed and agreed with the DMC** to ensure they understand the format of the data they will receive. The DMC can then confirm that the content proposed for the report is relevant to the DMC’s decision-making, clear, easy-to-follow, and easily interpretable.
  - Note: For some BR trials, no IS and IP is assigned; the DMC responsibilities are handled by the TS and TsP.
- **Blinded Dry-runs** are recommended for DMC deliverables; however, there is no need to import those Dry-run outputs into DMS.
- **DMC outputs / DMC Reports** should be loaded to DMS after DBL/unblinding of the study.
- For DMC **activities performed by a CRO**, the roles and responsibilities for reporting and document management will be done as described in the contract agreement.

DMC Process
Requests for additional analyses

If the DMC requires additional analyses during the DMC meeting, requests should be made directly to the IS/IP. The independent team is empowered to provide these analyses without informing CTT/Sponsor. Any departure from the previous analysis plan will be appropriately documented by the DMC members in the Closed Session minutes.

- If additional analyses / updates to already planned TFLs have **no unblinding implications**, updates should be made to the DMC SAP and TFL Shells document via an Amendment by the IS/IP and the outputs should be delivered to the DMC in future meetings.
- If additional analyses / updates to TFLs do **have unblinding implications**, these should be documented separately from the SAP by the IS/IP. **This document should not be shared with Novartis until final database lock.**

DMC Process
Use of interactive visualization tools

In some settings, an **interactive visualization tool (IVT)** can be used to **supplement** the static DMC Closed Report.

- Key example for an interactive visualization tools (IVT) is the (internal) **R Shiny DMC app**.
  - Note: If this app is to be used, teams need to evaluate if it is to be used “as is” or if it needs to be tailored to the study. If the latter, the App owner needs to be contacted to implement and validate (incl documentation).
- DMC members **decide whether to use an IVT** in the DMC closed session.
  - If the decision is made to use an IVT, capture the decision in the KOM meeting minutes.
  - If the decision is made after the DMC KOM, capture the decision in the open session DMC meeting minutes.
- DMC charter must **reflect that an app is being used** and include a description of the data-access control, and the end-to-end data flow related to the usage of the IVT.
- **IVTs may include unblinded data** and are currently hosted on Novartis access-controlled server.
  - Blinding/unblinding plan must reflect who has access to unblinded data, e.g., if someone from Novartis IT handles data as part of setting up the IVT, this must be mentioned in the blinding/unblinding plan.
- Those who get **access to the app** must be trained by the Trial Statistician.
  - Documentation of training needs to be archived in DMS.
- See **go/dmc/report** for more details on the use of IVT.

[Back to TOC](#)

Lesson 3: Other GxP Reporting Activities

NOVARTIS | Reimagining Medicine

Other GxP Reporting Activities

Purpose and Scope (1/2)

- **In scope** are analyses results intended to be used for a GxP Reporting Activity (RA), other than the CSR/IA/DMC, i.e., **RA for which documentation of the (process) validation / verification activities is required**. It is the **CTT responsibility** to assess whether such documentation is needed.
  - Assessment of the need for documentation should be based on the business impact and potential risks associated with the deliverables use. For example, for internal decision making by the Novartis enterprise governance boards at decision points, it is recommended to have complete RAP documentation.
  - Governing processes for these RAs are SOP-7017840 *Programming Development and Validation in Statistical Computing Environment (SCE)* and SOP-7012383 *Analyzing Clinical Data*.
- **Out-of-scope** are these RAs, as these are not governed by the above-mentioned SOPs:
  - Activities in scope of Pharmacometrics (PMX) Modeling Reports & Memos (refer to SOP-7014885: *Pharmacometrics Analysis of Clinical Data Supporting Interactions with Health Authorities).*
- Note: For the **CSPD activities,** please refer to go/CSPD_WBT which details the documentation requirements (SAP, TFL Shells, PDS, PDT, PSR) for submission summary documents, including the pooling strategy.

Other GxP Reporting Activities

Purpose and Scope (2/2)

Below are some **examples of RA** for which documentation of validation / verification activities is **required**:

- Abstracts / Posters / Conference Presentations
- Bioresearch Monitoring Program (BIMO)
- Briefing Books
- Companion Diagnostic (CDx) dossiers
- HA Requests
- Deliverables of studies with model-informed Dose Escalation
- Health Economics and Outcomes Research (HEOR) / Pricing and Reimbursement dossiers
- Internal decision-making (e.g., Transition Decision Point, Thorough QT/QTc (TQT) - waiver)

Note: It is possible that for an Exploratory analyses, the documentation of the validation/verification activities is required. Refer to the slide 62.

Other GxP Reporting Activities

Roles and Responsibilities

Once a request for RA deliverables has been made, Identify roles & responsibilities

- **Deliverables Lead (DL):** accountable and responsible for the set-up, control and **management of the programming activities** related to the production of deliverables within the SCE.
- **Deliverables Analyst (DA):** primarily accountable and responsible for providing the **analysis specifications** (e.g., Statistician, PMX Modeler, Biomarker Quantitative Specialist).
- **Deliverables Developer (DD):** in charge of the development of datasets and/or TFL outputs, i.e., programming skills are needed (e.g., Statistician, Data Scientist, Statistical Programmer).
- **Deliverables Validator (DV):** in charge of the validation, i.e., programming skills are needed (e.g., Statistician, Data Scientist, Statistical Programmer).

Identify documentation requirements before programming starts

- **Analysis plan** – available in DMS, as it is needed for the created the Validation plan. **Responsible role: DA**.
- **Validation plan** defines the validation strategy, whether is using programming verification (critical, most-critical) or a testing strategy. It is recommended to use the CSR-PSR template to document the plan. **Responsible role: DL, supported by the DA**.
- **PDT** documents the deliverables list to manage/track the production of these and their quality. It is recommended to use the CSR-PDT template to track programing activity. **Responsible role: DL**.
- **Note: For HA requests** and **trials with a Dose Escalation (DE) component**, the Validation plan may be developed in parallel with the programming activities.

Before release of final deliverables

- **Program verification: DD/DV** to ensure creation/validation of outputs per specifications.
- **Finalize specifications** (Analysis plan / TFL Shells, PDS (if applicable)) . **Responsible role: DA (Analysis Plan), DL (TFL Shells) (and DL for PDS, if applicable)**
- **Final run:** execute, verify and finalize the production of the requested deliverables. **Responsible role: DL.**
- **Release of deliverables: Responsible role: DL.**
- **PSR finalization: Responsible role: DL.**

Other GxP Reporting Activities

Required Documentation

Whether planned or unplanned, once it is agreed that **documentation of the validation/verification** is needed, the activities described in SOP-7017840 *Programming Development and Validation in SCE* and SOP-7012383 *Analyzing Clinical Data* should be adhered to. The following documentation is needed:

- **Specifications** for the data analyses and reporting: Analysis Plan / TFL Shells.
  - It is the team's decision to document the Analysis Plan / TFL Shells using one document or separate documents. Assess the type of analysis and/or reporting complexity to decide what is more appropriate. Note: There is no analysis plan template. See the Resources slide for Analysis Plan examples.
  - The DA is the **author** of the Analysis Plan; the DL is the author of the TFL Shells.
  - **Contributors** to the Analysis Plan / TFL Shells may include PKS Lead, Safety Lead, Biomarker Expert, PRO Expert, Imaging Expert, Modeler, Regulatory Affairs representative, Scientific Engagement & Communications representative.
  - **Analysis Plan / TFL Shells approvers**: LS, LP. Other approvers may be added as needed, depending on the purpose of the RA (e.g., ML / CD Lead, PKS Lead).
- **Technical specifications:** PDS (if needed), PDT, and PSR.
  - The initial version of the PSR defines the **Validation Plan.** See the PSR template for additional guidance.
- **Analysis specifications** (PDS, if needed) must be finalized and approved before the final run in the SCE.
- A **Dry-run** is recommended, especially if deliverables' recipients are external to Novartis.

Other GxP Reporting Activities

Required Documentation

It is recommended to have the **PSR approved by the LP within 1 week** after outputs are finalized in DMS.

- For unplanned analysis like exploratory, ad hoc analysis, HAQs, publications, etc., a project level PSR can be maintained in the SCE as a live document and updated (as new versions) for each subsequent analysis of a RA (as per the PSR template). The PSR is to be created and maintained at project UTIL in SCE but the final version should be in DMS for the approval process.
- There is a **PSR for every RA.**
- Refer to the **PSR template** for more details on the content, including instructions related to outsourced programming.

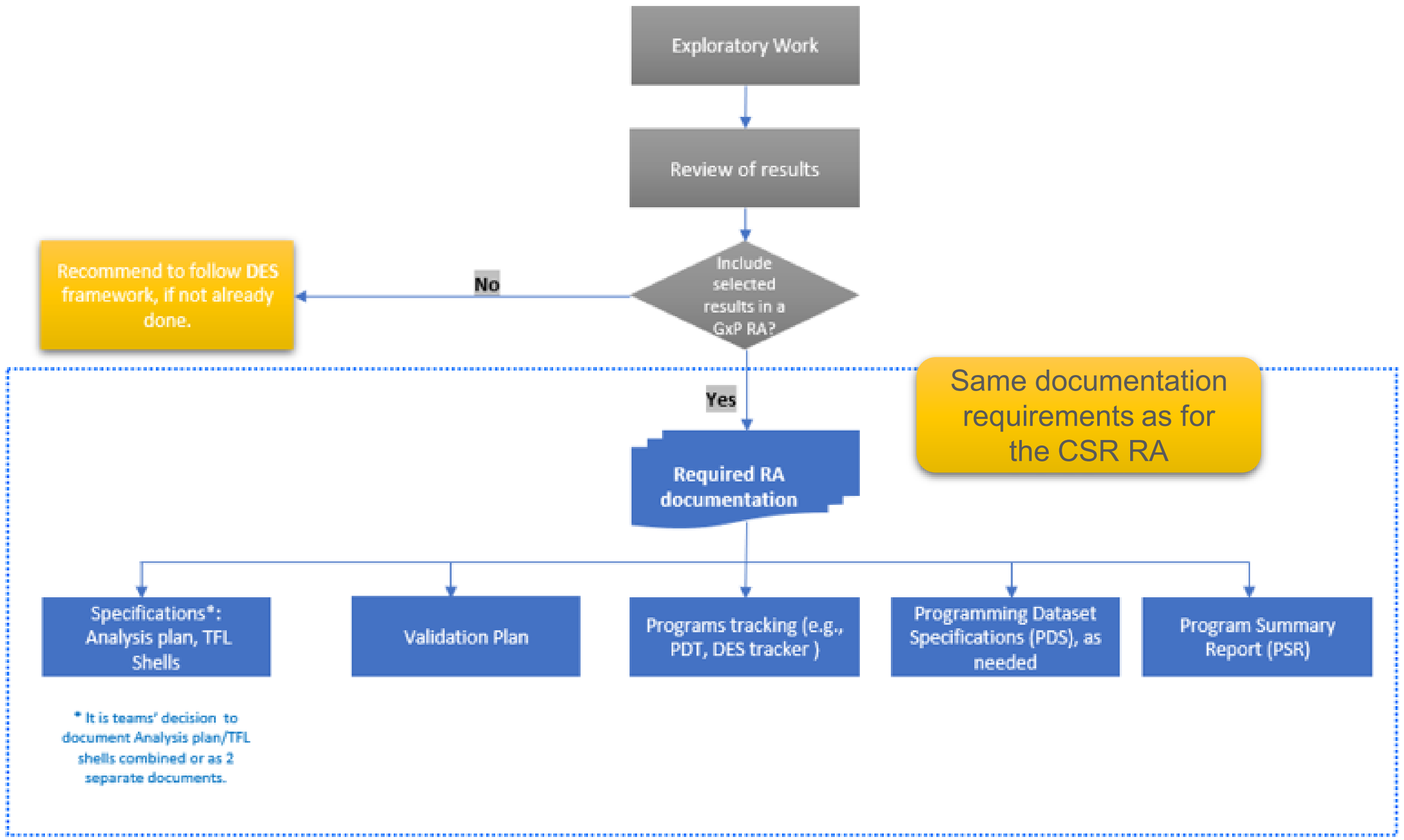
Other GxP Reporting Activities

Authors, Participants and Approvers

Document / Component	Author	Participants (Contributors / Reviewers)	Approvers
Analysis Plan	DA	**Contributors:** May include: PKS Lead, Safety Lead, Biomarker Expert, PRO Expert, Imaging Expert, Modeler, Regulatory Affairs rep, Scientific Engagement & Communications rep. **Reviewers**: LS, LP	LS, LP *Note:* Other approvers may be added based on the study specifics.
TFL Shells	DL	**Contributors:** May include: PKS Lead, Safety Lead, Biomarker Expert, PRO Expert, Imaging Expert, Modeler, Regulatory Affairs rep, Scientific Engagement & Communications rep. **Reviewers**: LS, LP	LS, LP *Note:* Other approvers may be added based on the study specifics.
PDS (SDTM & ADaM), if applicable	DL	**Reviewer:** DA	LP
PDT	DL	**Reviewer:** DA	*No Approval*
Validation Plan	DL	**Reviewer:** DA	LP

Other GxP Reporting Activities

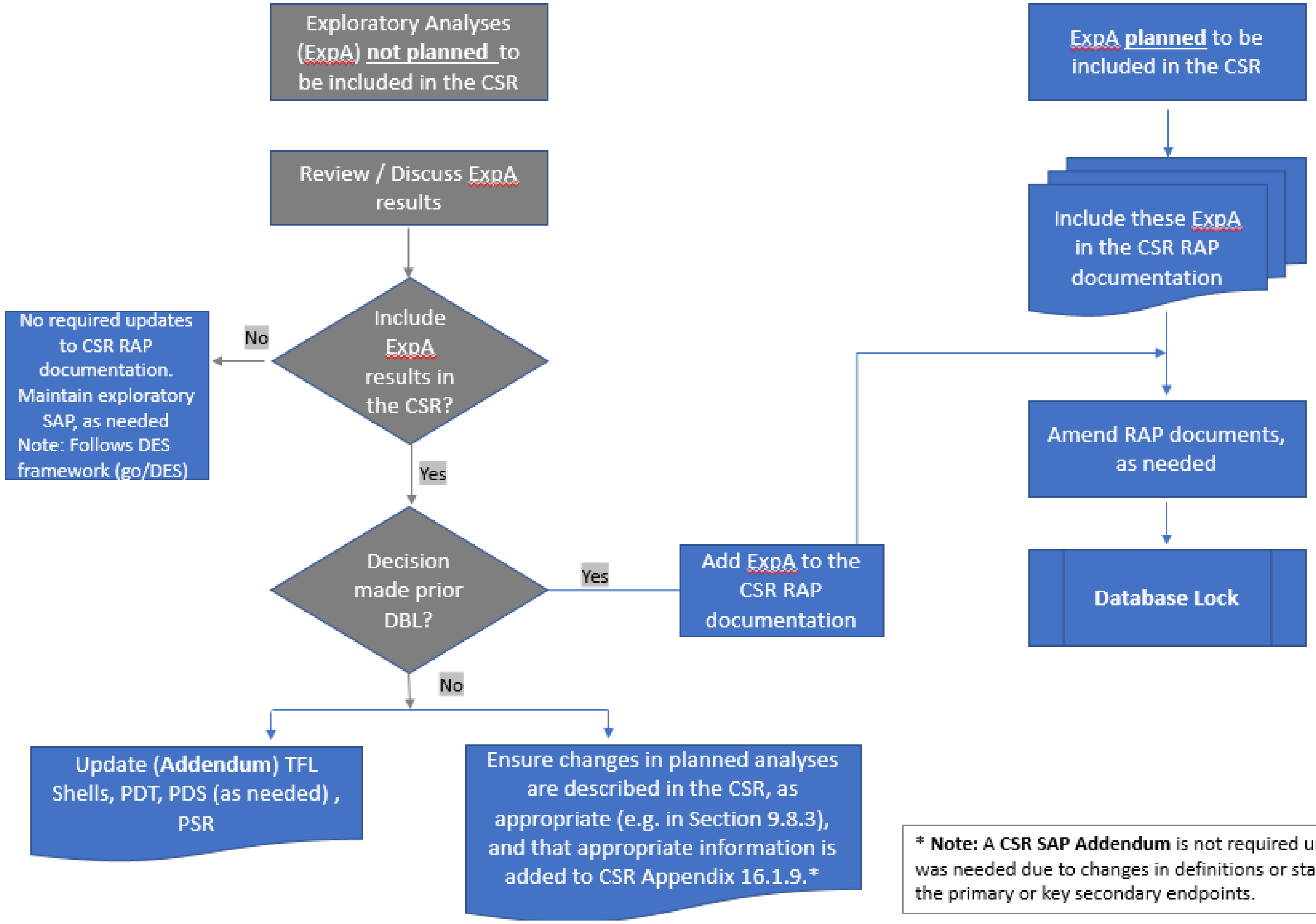
Required Documentation



Clarification note for GMA Non-CSR RA: 1) If TS is author, then LS should approve; 2) If LS is author, then DASL should approve; 3) If LS is author and the DASL, then LS can approve.

Other GxP Reporting Activities

Documentation of Exploratory Analyses



* **Note:** A **CSR SAP Addendum** is not required unless an Addendum was needed due to changes in definitions or statistical methods for the primary or key secondary endpoints.

Refer to go/Analytics_Exploratory in go/SMARThome for additional guidance

Other GxP Reporting Activities

Documentation of Exploratory Biomarker Analyses - Analysis Plan

Author	Contributors/Reviewers	Approvers in DMS
Exploratory Biomarker Data Scientist	• Exploratory Biomarker Programmer • Precision Medicine Lead • Exploratory Biomarker Programmer Lead • MW (if assigned) • PKS Lead (if PK is included in Exploratory Biomarker analysis)	• Exploratory Biomarker Data Scientist • Exploratory Biomarker Programmer Lead

Back to TOC

Appendix
Acknowledgements
References
Resources
Acronyms

Appendix - Example of SAPs Supporting Two CSRs

Protocol defines:
- primary analysis and key secondary endpoints at Wk32 (with DBL for primary efficacy analysis CSR*)
- final analysis at Wk64 (after final DBL; for final CSR)

Best Practice: Create **two SAPs** - one for each reporting event as aligned with GPS folder structure

Wk32 CSR SAP:
CABC123D4567_SAP_CSR_1

Wk32 CSR SAP needs updating:
CABC123D4567_SAP_CSR_1_amendment1

Wk32 DBL

- **Wk32 CSR SAP** needs updating for Wk32 CSR primary or key secondary endpoints: **CABC123D4567_SAP_CSR_1_addendum1**
- All other updates are:
  - ✓ added to CSR Appendix 16.1.9
  - ✓ added to TFL Shells
  - ✓ described in CSR (e.g., Section 9.8.3 'Changes in planned analyses')

Wk64 CSR SAP *(may include the Wk32 analyses or refer to the Wk32 SAP)*
CABC123D4567_SAP_CSR_2

Wk64 CSR SAP needs updating:
CABC123D4567_SAP_CSR_2_amendment1
(Note: There is no updating of primary and key secondary endpoints as they have been reported in the Wk32 CSR.)

Wk64 DBL

- Document changes to Wk64 DBL planned analysis in the **CSR Appendix 16.1.9,**
- TFL Shells update as addendum
- Describe, high levels, changes and rationale in CSR Section 9.8.3 'Changes in planned analyses')

Note: * This SAP is submitted to Registries, if required.

Acknowledgments

- **Sponsor** - Steffen Ballerstedt
- **RAP WBT Team Leads** - Maritza Melendez, Ayan Das Gupta
- **RAP WBT SME** – Maria-Athina Altzerinakou, Chandrasekhar Bhupathi, Tiina Kirsilae, Prasanna Kumar Nidamarthy, Janardhana Vemula, Deepak Yeole, Danjie Zhao
- **Focused Contributors/SMEs**
  - CDx Bridging Studies: Jiawei Duan, Alexandra Vaury
  - CP: Michelle Quinlan
  - DMC: David Lawrence, Tobias Muetze
  - DO&A QA: Shiju Narayanamoorthy
  - Outsourcing: Michelle Quinlan, Jackie Han
  - RWS: Daniel Strausak, Leila Virkki
- **Facilitator:** Linda Kanitra

References

- **SOP-7012383**: *Analyzing Clinical Data*
- **SOP-7017840:** *Programming Development and Validation in Statistical Computing Environment (SCE)*
  - **WP-7001131**: *Programming Development and Validation Instructions*
  - **GUID-8160112**: *Best Practices for Programming Development and Validation*
- **SOP-7018147**: *GxP document Lifecycle Management*
- **SOP-8109242**: *Clinical Study Report*
- **WP-7007641:** *JPTO_JOM-00105_Guidance for how to handle Japan specific outputs for Trial/Project Level*
- Note: See slide 47 for DMC related References.

Resources

- CDO - go/CDO
- Clinical Trials Safety Disclosure - go/CTSD
- Clinical Pharmacology – go/CP
- CSR Guidance – go/CSR_Guidance
- eCRS - go/eCRStraining
- DEM - go/DEM
- DMC - go/DMC
- Estimands - go/Estimands
- Global Standard TFL Library – go/Global_Standard_TFL_Library
- Guidances for AQS Biostats - go/AQS_Guidances
- GxP RA Analysis Plan examples – go/Non-CSR Request
- ICH – go/ICH
- Introduction to PDS Tool and Process - go/PDS
- International Research and Development Document Standards (IRDDS) Training - go/IRDDS
- Novartis Clinical Vault AQS 1-Stop-Shop – go/AQS_NCV
- Outsourcing – go/AQS_Outsourcing
- Protocol template - go/OneCTP
- PSR template
- Randomization Office – go/Randomization_Office
- RAP additional supporting materials - go/PRADa
- SAP template – go/AQS_SAP
- SMART training – go/SMARThome

Acronyms – A to D (1/3)

ADaM	Analysis Dataset Model
AE	Adverse Event
AQS	Advanced Quantitative Sciences
BIMO	Bioresearch Monitoring
BR	BioMedical Research
CD	Clinical Development
CD Lead	Clinical Development Lead
CDC	Clinical Data Complete
CDO	Clinical Disclosure Office
CDx	Companion Diagnostic
CP	Clinical Pharmacology
CPP	Clinical Pharmacology Partnership
CRF	Case Report Form (and eCRF)
CRO	Contract Research Organization
CSPD	Clinical Summary Preparation Documentation
CSR	Clinical Study Report
CT	Clinical Trials
CTIS	Clinical Trials Information System

CTP	Clinical Trial Protocol
CTSD	Clinical Trial Safety Disclosure
CTT	Clinical Trial Team
DA	Deliverables Analyst
DASL	Disease Area Statistics Lead
DBL	Database Lock
DD	Deliverables Developer
DE	Dose Escalation
DEM	Dose Escalation Meeting
DES	Data Exploration Strategy
DL	Deliverables Lead
DM	Data Manager/Data Management
DMC	Data Monitoring Committee
DMS	Document Management System
DO	Data Operations
DO&A QA	Data Operations and Analytics Quality Assurance
DQP/DQT	Data Quality Plan/Team
DR	Dry-run
DSMB	Data Safety Monitoring Board
DSUR	Development Safety Update Report

Acronyms – D to P (2/3)

DTS	Data Transfer Specifications
DU	Development Unit
DV	Deliverables Validator
ED-AQS	Early Development - Advanced Quantitative Sciences
ESOPS	Electronic Standard Operating Procedure System
eRAP	easy RAP Tool
EU	European Union
FDA	Food and Drug Administration
FIR	First Interpretable Results
FPFV	First Patient First Visit
GCO	Global Clinical Operations
GGH	Global Group Head
GMA	Global Medical Affairs
GPBH	Global Program Biostatistics Head
GPS	Global Programming and Statistical Environment
GUID	Guidance Document
HA	Health Authority / Health Authorities
HEOR	Health Economics and Outcomes Research
IA	Interim Analysis
IB	Investigator's Brochure

ICH	International Council for Harmonisation
iDMC	Independent Data Monitoring Committee
IP	Independent Programmer
IS	Independent Statistician
IVT	Interactive Visualization Tool
KOM	Kick off Meeting
LF	Line Function
LP	Lead Programmer
LPLV	Last Patient Last Visit
LS	Lead Statistician
MAA	Marketing Authorization Application
ML	Medical Lead
MW	Medical Writer
NCV	Novartis Clinical Vault
NDA	New Drug Application
One CTP	One Common Technical Protocol
PD	Pharmacodynamics
PDS	Programming Datasets Specifications
PDT	Programming Deliverables Tracker
PK	Pharmacokinetics
PKS	Pharmacokinetic Sciences
PMX	Pharmacometrics
PP	Project/Program Programmer
PRO	Patient Reported Outcomes
PS	Project Statistician
PSR	Programming Summary Report
PSUR	Periodic Safety Update Report

Acronyms – R to X (3/3)

RA	Regulatory Affairs
RA	Reporting Activities
RAP	Reporting and Analysis Process
RMP	Risk Management Plan
RO	Randomization Office
SAP	Statistical Analysis Plan
SAS	Statistical Analysis System
SBP	Summary of Biopharmaceutics and Associated Analytical Methods
SCE	Statistical Computing Environment
SCE	Summary of Clinical Efficacy
SCP	Summary of Clinical Pharmacology
SCS	Summary of Clinical Safety
SDTM	Study Data Tabulation Model
SL	Study Lead
SMART	Streamlined Analysis Reporting & Advanced Exploration
SME	Subject Matter Expert

SOP	Standard Operating Procedure
SP	Statistical Programming
SSW	Study Specific Worksheet
TA	Therapeutic Area
TCO	Translational Clinical Oncology
TFL	Table, Figures and Listings
TM	Translational Medicine
TMF	Trial Master File
TOC	Table of Contents
TsP	Trial Statistical Programmer
TQT	Thorough QT/QTc
TS	Trial Statistician
WBT	Web Based Training
WP	Working Practice
XML	Extensible Markup Language